

Causes of primary adrenal insufficiency (Addison disease)

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Literature review current through: **Jun 2026**.
This topic last updated: **Jun 18, 2026**.

INTRODUCTION

When Thomas Addison described the disease that now bears his name [1], bilateral [adrenal](#) destruction by [tuberculosis](#) was its most common cause. Now [tuberculosis](#) accounts for only 7 to 20 percent of cases; autoimmune disease is responsible for 70 to 90 percent, with the remainder being caused by other infectious diseases, replacement by [metastatic cancer](#) or [lymphoma](#), [adrenal hemorrhage](#) or [infarction](#), or drugs ([table 1](#)) [2,3]. Disseminated tuberculous or fungal infections are still a major cause of [adrenal insufficiency](#) in populations with a high [prevalence](#) of these diseases, but as [tuberculosis](#) has been better controlled, the overall [incidence](#) of [Addison disease](#) has decreased [3,4]. The [prevalence](#) of [Addison disease](#) in Western countries has been estimated at 35 to 60 per million, but three studies indicate it may be as high as 210 per million [5-8].

This topic will review the major causes of [primary adrenal insufficiency](#). The pathogenesis, clinical manifestations, diagnosis, and treatment of [adrenal insufficiency](#) are reviewed separately. Other rare causes of [adrenal insufficiency](#), such as [adrenoleukodystrophy/adrenal myeloneuropathy](#), inherited disorders that present in [infancy](#) or childhood are also reviewed separately ([table 1](#)).

- (See "[Clinical manifestations of adrenal insufficiency in adults](#)".)
- (See "[Determining the etiology of adrenal insufficiency in adults](#)".)
- (See "[Treatment of adrenal insufficiency in adults](#)".)
- (See "[Clinical features, evaluation, and diagnosis of X-linked adrenoleukodystrophy](#)".)

AUTOIMMUNE ADRENALITIS

Pathogenesis — What long was termed "idiopathic" [primary adrenal insufficiency](#) is the result of an autoimmune process that destroys the [adrenal cortex](#). There is evidence of both [humoral](#) and cell-mediated immune mechanisms directed at the [adrenal cortex](#), often associated with autoimmune destruction of other endocrine glands. (See '[Polyglandular autoimmune syndromes](#)' below.)

Patients who develop isolated autoimmune [adrenal insufficiency](#) during the first two decades of life are predominately male (71 percent), whereas those diagnosed during the third decade are equally male and female, and those diagnosed thereafter are predominantly female (81 percent) [9]. The reason for these sex differences is unknown.

- **Humoral immunity** – [Autoantibodies](#) that react with several steroidogenic enzymes (most often [21-hydroxylase](#)) and all three zones of the [adrenal cortex](#) are present in the serum of up to 86 percent of patients with autoimmune [primary adrenal insufficiency](#) ([figure 1](#)), but only rarely in patients with other causes of [adrenal insufficiency](#), or in healthy individuals [7,10]. However, up to 10 percent of first-degree relatives of patients with autoimmune [primary adrenal insufficiency](#) express these [autoantibodies](#) and have an increased risk of developing [adrenal](#)

Genetics – The *CYP21A1P* (P450c21A) and *CYP21A2* (P450c21B) genes are located in the human leukocyte antigen (HLA) class III region on [chromosome 6](#) [11]. This co-localization underlies the associations between certain [HLA alleles](#) and risk of autoimmune [adrenal insufficiency](#).

- **Susceptibility alleles** – Genetic susceptibility to autoimmune [adrenal insufficiency](#) is strongly associated with [HLA B8](#), [DR3](#), and [DR4 alleles](#), except when it occurs as part of [polyglandular autoimmune syndrome](#) (PAS) type 1, in which no [HLA](#) association has been found ([table 2](#)) [12-14]. Several specific [HLA](#) class II antigens, including DQA1*0301, DQA1*0501, DQB1*0201,

Immune and genetic features of autoimmune adrenal insufficiency			
Immune or genetic feature	Polyglandular autoimmune syndrome type 1 (PA1/SAPS)	Isolated autoimmune adrenal insufficiency or polyglandular autoimmune syndrome type 2	
Presence of autoantibodies to:			
CYP11A2	50 to 60%	<10%	
CYP11B1	50 to 50%	<10%	
CYP11A1	50 to 50%	50 to 80%	
Associated HLA genotypes			
Increased risk of AI	HLA-B8, DR3, and DR4 alleles	HLA-B*0801, DR3, and DR4 alleles	

DQB1*0302, DRB1*0404, and DRB1*0301, are associated with sporadic autoimmune adrenal insufficiency and PAS type 2 [15-18]. Alterations in the HLA class I chain-related microsatellite also confer susceptibility [19]. (See 'Polyglandular autoimmune syndromes' below.)

		* DRB1*0301 * DRB1*0302 * DRB1*0404 * DRB1*0301 * DRB1*0302 * DRB1*0404	
Increased risk of Addison's disease	None	Associated with Addison's disease	
HLA-A*01:01	No	No	
HLA-B*08:01	No	No	
HLA-C*04:01	No	No	
HLA-D*01:01	No	No	
HLA-E*01:01	No	No	
HLA-F*01:01	No	No	
HLA-G*01:01	No	No	
HLA-I*01:01	No	No	
HLA-J*01:01	No	No	
HLA-K*01:01	No	No	
HLA-L*01:01	No	No	
HLA-M*01:01	No	No	
HLA-N*01:01	No	No	
HLA-O*01:01	No	No	
HLA-P*01:01	No	No	
HLA-Q*01:01	No	No	
HLA-R*01:01	No	No	
HLA-S*01:01	No	No	
HLA-T*01:01	No	No	
HLA-U*01:01	No	No	
HLA-V*01:01	No	No	
HLA-W*01:01	No	No	
HLA-X*01:01	No	No	
HLA-Y*01:01	No	No	
HLA-Z*01:01	No	No	

Immune and genetic features of autoimmune adrenal insufficiency

Table 2 - larger image below

- **Protective alleles** – Certain HLA alleles may protect against the development of adrenal insufficiency. In one study of 118 individuals with 21-hydroxylase autoantibodies and the HLA DR4 allele, patients with confirmed adrenal insufficiency were much less likely to carry the HLA B15 allele (1 of 55 [2 percent]) compared with those who did not have adrenal insufficiency (24 of 63 [40 percent]) [20]. On follow-up, none of those with the HLA B15 allele developed adrenal insufficiency, whereas 25 percent of those without the HLA B15 allele progressed.

Initial laboratory findings — The first evidence of autoimmune adrenal insufficiency is usually an increase in plasma renin activity in association with a normal or low serum aldosterone concentration, suggesting that the zona glomerulosa is involved initially [21,22]. Several months to years later, zona fasciculata dysfunction becomes evident, first by a decreasing serum cortisol response to corticotropin (ACTH) stimulation, later by increased basal serum ACTH concentrations, and finally by decreasing basal serum cortisol concentrations with symptoms of cortisol deficiency [21,23].

In one study of patients with known CYP21A2 antibodies but without clinical cortisol deficiency, an elevated serum ACTH best predicted progression to frank adrenal insufficiency. Plasma renin activity and peak cortisol response to exogenous ACTH did not help to predict progression [24].

Polyglandular autoimmune syndromes — The combination of autoimmune adrenal insufficiency with other autoimmune endocrine and nonendocrine disorders has been referred to as the PAS types 1 and 2 [12,25-27]. Approximately 50 to 65 percent of patients with autoimmune adrenal insufficiency have one or more other autoimmune endocrine disorders (table 3) [7,28-33]. On the other hand, patients with the more common autoimmune endocrine disorders (eg, type 1 diabetes, chronic autoimmune thyroiditis, or Graves' disease) rarely develop adrenal insufficiency.

Management of PAS types 1 and 2 involves treating the individual autoimmune disorders, reviewed in detail separately.

- (See "Treatment of adrenal insufficiency in adults".)
- (See "Adrenal insufficiency in children: Treatment".)
- (See "Chronic mucocutaneous candidiasis", section on 'Treatment'.)
- (See "Hypoparathyroidism", section on 'Management'.)
- (See "Treatment of primary hypothyroidism in adults".)
- (See "Hypothyroidism in children and adolescents: Clinical features, diagnosis, and management", section on 'Treatment and prognosis'.)

Type 1 (monogenic) — PAS type 1, also referred to as the autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED) syndrome, is a rare autosomal recessive disorder. The clinical diagnosis is established by the presence of ≥2 of the following:

- **Hypoparathyroidism** – The hypoparathyroidism may [34,35] or may not [36] occur in association with antiparathyroid gland antibodies that are directed against the calcium-sensing receptor. (See "Etiology of hypocalcemia in adults", section on 'Autoimmune'.)
- **Chronic mucocutaneous candidiasis** – The candidiasis almost always involves the mouth, although it may involve just the nail beds or be more extensive (picture 1) [25]. It is chronic or recurrent, and it is resistant to conventional therapy. Oral mucous squamous cell carcinoma has been rarely reported in association with the candidiasis of PAS type 1 [37]. (See "Overview of Candida infections", section on 'Chronic mucocutaneous candidiasis'.)
- **Adrenal insufficiency** – In patients with PAS type 1, CYP21A2 autoantibodies are most commonly detected (table 2), followed by CYP11A1 or CYP17 autoantibodies [38,39]. Patients with autoantibodies develop adrenal insufficiency at a rate of up to 19 percent per year [21,23,40], and, in PAS type 1, the presence of adrenal autoantibodies strongly predicts the development of adrenal insufficiency (eg, >90 percent positive predictive value) [23,36,41]. Younger age is associated with a higher prevalence of autoantibodies and greater predictive value for developing adrenal insufficiency [42-44].

Females are affected slightly more frequently than males [12,25,26], and PAS type 1 is most common among Finns, Sardinians, and Iranian Jews; sporadic cases have also been identified elsewhere, including most European countries [45,46].

Genetic basis — The classic clinical presentation of PAS type 1 is an autosomal recessive disorder caused by pathogenic variants in the autoimmune regulator (AIRE) gene on chromosome 21q22.3 [47-49]. The predominant genetic variant differs across patient populations [49-

[52]. There is no apparent correlation between specific pathogenic variants and phenotype. The variable clinical presentation and progression are presumably due to environmental factors and genetic factors other than *AIRE* variants [49].

An atypical presentation with autosomal dominant inheritance has been designated as "nonclassic" PAS type 1. Dominant-negative mono-allelic variants in the first zinc finger domain of the *AIRE* gene result in a later presentation with fewer clinical features [53].

The *AIRE* gene product is expressed in the thymus, lymph nodes, pancreas, adrenal cortex, and fetal liver; it functions as a nuclear transcription factor and induces a variety of tissue-specific antigens in thymic medullary epithelial cells. T-cells that react to these self-antigens are then deleted, inducing "tolerance" to self-antigens [54]. In the absence of a functional *AIRE* gene, autoantibodies may be generated [55].

Clinical presentation

• **Classic presentation** – In the classic presentation of PAS type 1, hypoparathyroidism or chronic mucocutaneous candidiasis is usually the first manifestation, characteristically appearing during childhood or early adolescence, always by the early twenties (table 4) [9,12,25,26]. Adrenal insufficiency usually develops later, at age 10 to 15 years. The antigen targets are adrenal enzymes including P450scc (side-chain cleavage enzyme), P450c17 (17-alpha-hydroxylase), and P450c21 (21-hydroxylase) [56].

Other features may include the following:

- Primary hypogonadism occurs in approximately 60 percent of patients. (See "Autoimmune primary ovarian insufficiency (premature ovarian failure)".)
- Malabsorption and other gastrointestinal disorders occur in approximately 25 percent of patients [57,58], 90 percent of whom have autoantibodies to tryptophan hydroxylase, an intestinal antigen. Only one-third of patients without gastrointestinal disorders have these autoantibodies. One report described a patient with severe malabsorption syndrome due to a deficiency of cholecystokinin-producing enteroendocrine cells in the mucosa of the proximal small bowel [59].
- Enamel hypoplasia is also common [60].
- Autoimmune hepatitis was reported in a series of Finnish and Swedish patients [61].

Type 1 diabetes and chronic autoimmune thyroiditis are uncommon. Graves' disease is not associated with this disorder (table 4). In one series of 68 Finnish patients, approximately 50 percent of patients developed the triad of candidiasis, hypoparathyroidism, and adrenal insufficiency [25]. The incidence of other manifestations differed from that reported in other smaller series and reviews of the literature [25]. These differences from previous reports may reflect a variant form of the disorder or a more thorough examination of affected patients. For example, autoimmune diabetes was 12 times more prevalent, and manifestations of ectodermal dysplasia, including dental enamel hypoplasia, pitted dystrophy of the nails, keratopathy, and calcified plaques on the tympanic membranes, were present in one-third to two-thirds of the patients.

• **Nonclassic presentation** – These patients have pathogenic variants in *AIRE* and may present with classic PAS type 1 manifestations as described above, but with onset during adulthood, or they may present with PAS type 2, or isolated, organ-specific autoimmunity (especially vitiligo and pernicious anemia) [53,62,63]. In a study of 28 patients, "minor" features appeared first, including urticarial eruption (78.6 percent), intestinal dysfunction (53.6 percent), and enamel hypoplasia (42.9 percent) [62]. The variable clinical presentation and progression are presumably due to environmental factors and genetic factors other than *AIRE* variants [49].

Type 2 (polygenic) — PAS type 2 is much more prevalent than type 1. This syndrome entails any presentation of polyglandular autoimmunity that includes adrenal insufficiency but **not** hypoparathyroidism or chronic mucocutaneous candidiasis (ie, PAS type 1). Many combinations of overlapping endocrine and nonendocrine autoimmune presentations are possible. Initial nomenclature that defined multiple subtypes has been abandoned, and all non-APECED presentations are referred to as type 2. Approximately one-half of cases are familial, and several modes of inheritance (autosomal recessive, autosomal dominant, and polygenic) have been reported [9,12,26,64]. Female adults are affected up to three times more often than male adults [65].

Clinical presentation — Primary adrenal insufficiency is the principal manifestation of PAS type 2 [12,26,65]. It occurs later than in type 1; the age of onset ranges from childhood to late adulthood, with most cases occurring between the ages of 20 and 40 years [9,12,32,65]. Autoimmune thyroid disease (usually chronic autoimmune thyroiditis, but occasionally Graves' disease) and type 1 diabetes are also common (table 3 and table 4). Hypoparathyroidism does **not** occur in this disorder, and alopecia and pernicious anemia are much less frequent than in the type 1 syndrome.

Adrenal insufficiency is the initial manifestation in approximately 50 percent of patients, occurs simultaneously with autoimmune thyroid disease or diabetes mellitus in approximately 20

Types of endocrine and nonendocrine autoimmune syndromes associated with adrenal insufficiency	
Disorder	Prevalence (%)
Polyglandular autoimmune syndrome type 1	
<i>Endocrine</i>	
Hypoparathyroidism	88
Chronic mucocutaneous candidiasis	88
Adrenal insufficiency	88
Primary hypogonadism	88
Hypothyroidism	88
Type 1 diabetes mellitus	11
Hypoadrenalism	11
Hypothalamic hypopituitarism	11
<i>Nonendocrine</i>	
Hypoparathyroidism	25
Hypothyroidism	25
Hypogonadism	25
Chronic active hepatitis	9
Vitiligo	9
Polyglandular autoimmune syndrome type 2	
<i>Endocrine</i>	
Adrenal insufficiency	100
Autoimmune thyroid disease	50
Type 1 diabetes mellitus	50
Primary hypogonadism	5 to 10
Hypoparathyroidism	11
<i>Nonendocrine</i>	
Hypothyroidism	11
Hypogonadism	11
Hypoparathyroidism	11
Hypoadrenalism	11
Hypothalamic hypopituitarism	11
Hypoparathyroidism	11
Hypothyroidism	11
Hypogonadism	11
Hypoparathyroidism	11
Hypoadrenalism	11
Hypothalamic hypopituitarism	11

Types of endocrine and nonendocrine autoimmune syndromes associated with adrenal insufficiency

Table 4 - larger image below

percent, and follows them in approximately 30 percent [9,12,32]. In one series, however, adrenal insufficiency was the initial manifestation in only 19 percent of patients [65]. Primary hypogonadism can alternatively occur first and, as in PAS type 1, ovarian insufficiency is more frequent than testicular insufficiency [12,26,32]. (See "[Autoimmune primary ovarian insufficiency \(premature ovarian failure\)](#)".)

Hypopituitarism may occur due to autoimmune hypophysitis, which preferentially causes ACTH deficiency, but also may lead to thyrotropin or, rarely, growth hormone deficiency [66]. Other nonendocrine autoimmune disorders, such as vitiligo, myasthenia gravis, thrombocytopenic purpura, Sjögren's disease, rheumatoid arthritis, and primary antiphospholipid syndrome, occur occasionally [12,26,67], as does serositis with pericardial and/or pleural involvement [68]. Patients with vitiligo, as an example, can have autoantibodies to tyrosinase, an enzyme involved in the synthesis of melanin; tyrosinase-related proteins; or to a melanocyte-specific protein called Pmel17 [69]. (See "[Causes of hypopituitarism](#)" and "[Clinical manifestations of antiphospholipid syndrome](#)".)

Testing for adrenal insufficiency — Some experts endorse measuring 21-hydroxylase autoantibodies in all patients with type 1 diabetes, given the potential lethality of unrecognized adrenal insufficiency. However, in patients with type 1 diabetes, the prevalence of subnormal responses to ACTH is low (approximately 1 percent) ([table 5](#)); therefore, we do not perform broad testing in this population and instead monitor patients clinically for signs and symptoms of adrenal insufficiency. In one report, for example, 11 of 629 patients (1.7 percent) with type 1 diabetes had autoantibodies directed against 21-hydroxylase [70]. Of these 11 patients, three had adrenal insufficiency, five did not, and three had not been tested. In another study, only 1 of 114 children with type 1 diabetes and none of 35 children with autoimmune thyroiditis tested positive for anti-adrenal antibodies measured by immunofluorescence [71].

In people with type 1 diabetes and positive 21-hydroxylase autoantibodies, the risk of developing autoimmune adrenal insufficiency varies with genetic background. For example, among autoantibody-positive patients with both the diabetes-associated HLA DQ8 allele and the DRB*0404 allele, approximately 80 percent develop adrenal insufficiency [15,72]. In another report of 38 patients with type 1 diabetes and 21-hydroxylase autoantibodies, those who were homozygous for MICA5.1 (an MHC class I-related molecule located between tumor necrosis factor- α and HLA-B) were at higher risk for developing adrenal insufficiency than those without homozygosity [72,73].

In studies conducted in Scandinavia, patients with celiac disease had an 11-fold increased risk for adrenal insufficiency, and 6 of 76 patients with adrenal insufficiency had celiac disease, suggesting that testing for the other disorder is reasonable in patients with either one [74,75].

INFECTIOUS ADRENALITIS

A variety of infectious agents can infect the adrenal gland and lead to adrenal insufficiency.

Tuberculosis — Tuberculosis can destroy the adrenal glands. Tuberculous adrenalitis results from hematogenous spread from active infection elsewhere in the body [76]. Extraadrenal tuberculosis is usually evident, but may be clinically latent [76,77]. (See "[Miliary tuberculosis](#)", section on '[Clinical manifestations](#)'.)

Adrenal destruction is gradual, with the medulla being destroyed more often than the cortex, for unknown reasons [76]. Antiadrenal autoantibodies are **not** present in serum [78]. The adrenal glands are usually enlarged by inflammatory cell infiltration of the cortex and granulomas early in the disease; the increase in size can be detected by computed tomography (CT) or magnetic resonance imaging (MRI) [77,79,80]. Caseous nodules and fibrosis then gradually replace the adrenal gland tissue [77], so that after approximately two years, the adrenals become normal or small in size [77,79-81].

- Adrenal calcifications can be seen radiographically in 50 percent of patients [77,79,81]. However, the absence of enlarged or calcified adrenal glands does not rule out tuberculosis as the cause of adrenal insufficiency.
- Recovery of normal adrenal function may occur after effective antituberculous therapy [82], but usually does not [83].

The adrenal glands may be enlarged in patients with pulmonary tuberculosis even when adrenal function is normal. Adrenal size becomes smaller after successful treatment of the tuberculosis [84].

Disseminated fungal infections — Several species of fungi can involve the adrenal glands and cause adrenal insufficiency. Histoplasmosis [85-87] and paracoccidioidomycosis (South American blastomycosis) [88,89] are important causes of adrenal insufficiency in endemic areas. In contrast, adrenal insufficiency is rare in patients with cryptococcosis, coccidioidomycosis, and North American blastomycosis [90-92]. (See "[Pathogenesis and clinical manifestations of disseminated histoplasmosis](#)", section on '[Adrenal involvement](#)'.)

- The adrenal glands are enlarged and may become calcified in all of these disorders.

- Recovery of adrenal function after prolonged anti-fungal treatment can occur [86].

HIV infection — In the early acquired immunodeficiency syndrome (AIDS) epidemic, the diverse endocrine manifestations of human immunodeficiency virus (HIV) infection, including adrenal insufficiency, were more often a consequence of opportunistic infections, neoplasms, or concomitant systemic illness. The widespread use of potent antiretroviral therapy has led to a decline in the incidence of endocrine complications. (See "[Overview of endocrine dysfunction in patients with HIV](#)", section on 'Adrenal gland dysfunction'.)

Other infections — A few other infections are rare causes of primary adrenal insufficiency. These include:

- Syphilis, in which gumma formation and demonstrable spirochetes may be seen [76].
- African trypanosomiasis, unrelated to its treatment with [suramin](#), which can impair adrenal function when given in higher than the usual therapeutic doses [93]. In another study of 60 patients, five had insufficient cortisol responses to ACTH stimulation on admission, and another five had an abnormal response on discharge; all improved during follow-up evaluation [94].

HEMORRHAGIC INFARCTION

Acute adrenal insufficiency may occur as a result of bilateral adrenal infarction caused by hemorrhage or adrenal vein thrombosis [95-97]. Bilateral adrenal hemorrhages are present in approximately 1 percent of routine autopsies, and before the availability of CT scanning ([image 1](#)) and MRI, the diagnosis was usually made at autopsy [95,97].

- **Clinical presentation** – Evidence of occult hemorrhage, such as a sudden fall in hemoglobin and hematocrit, and progressive hyperkalemia, hyponatremia, and volume contraction, suggests the diagnosis. However, the condition is difficult to recognize clinically, and, despite the dramatic symptoms, the diagnosis is often missed. Without appropriate therapy, shock progresses to coma and death.

The symptoms and signs include hypotension or shock (>90 percent of patients); abdominal, back, flank, or lower chest pain (86 percent); fever (66 percent); anorexia, nausea, or vomiting (47 percent); confusion or disorientation (42 percent); and abdominal rigidity or rebound (22 percent) [96]. (See '[Bilateral adrenal injury](#)' below.)

- **Risk factors** – Major risk factors for adrenal hemorrhage include anticoagulant drug or heparin therapy (heparin-induced thrombocytopenia resulting in bilateral adrenal hemorrhage) [98], thromboembolic disease, hypercoagulable states such as antiphospholipid syndrome [99,100], physical trauma, the postoperative state, sepsis, and any cause of severe stress [97,101]. In patients treated with anticoagulants, clotting test results are usually within the therapeutic range, and spontaneous bleeding elsewhere is not evident [96].

Adrenal hemorrhage and death have been associated with meningococcemia (Waterhouse-Friderichsen syndrome) [102]. Waterhouse-Friderichsen syndrome should be considered in a patient with fever and petechiae, which are present in approximately 50 to 60 percent of patients [103]. Sepsis, bilateral adrenal hemorrhage, and death have also been reported with *Haemophilus influenzae*, *Escherichia coli*, *Mycoplasma pneumoniae*, *Streptococcus pneumoniae*, and *Staphylococcus aureus* infection, as well as with infective endocarditis, Rocky Mountain spotted fever, murine and epidemic typhus, and viral infections (including cytomegalovirus, parvovirus B19, and Epstein-Barr virus) [104-106]. In one report of 51 children dying of sepsis and bilateral adrenal hemorrhage, *Pseudomonas aeruginosa* was the most common pathogen [104]. (See "[Clinical manifestations of Staphylococcus aureus infection in adults](#)".)

- **Pathogenesis** – The pathogenesis of adrenal hemorrhage is unclear. Increased adrenal blood flow stimulated by corticotropin (ACTH) secreted in response to stress may play a contributory role [97]. Anticoagulation therapy is implicated in approximately one-third of patients, but adrenal hemorrhage occurs very rarely in patients who are anticoagulated; when it does, it is usually within the first 2 to 12 days of therapy [95,97,98]. A case-control study (23 patients with bilateral massive adrenal hemorrhage and 92 control patients) reported that thrombocytopenia, heparin use, and sepsis were the variables that were most strongly and independently associated with adrenal hemorrhage risk [101].

BILATERAL ADRENAL INJURY

Adrenal insufficiency is a potential complication of blunt trauma to the lower torso, including motor vehicle accidents [107]; cases have been recognized in the intensive care setting as a result of admission CT examinations [108].

Adrenal crisis can also occur as a result of sudden, bilateral adrenal necrosis caused by hemorrhage, emboli, sepsis, or, very rarely, adrenal vein thrombosis after a back injury [95,96,109]. These patients do not have evidence of preexisting adrenal insufficiency. Before CT became widely available, the diagnosis of adrenal hemorrhage was usually made at autopsy [95]. (See '[Hemorrhagic infarction](#)' above.)

- **Presenting symptoms** – The presenting symptoms and signs of bilateral adrenal hemorrhage (and the frequency with which they occurred in one report) include [96]:

- Hypotension or shock (>90 percent)
- Abdominal, flank, back, or lower chest pain (86 percent)
- Fever (66 percent, presumably a response to inflammation)
- Anorexia, nausea, or vomiting (47 percent)
- Neuropsychiatric symptoms such as confusion or disorientation (42 percent)
- Abdominal rigidity or rebound tenderness (22 percent)

Surprisingly, only approximately one-half of patients have hypotension before shock. The acute onset does not permit enough time for the patient to become hyperpigmented.

Evidence of occult hemorrhage, such as a sudden fall in hemoglobin and hematocrit, with progressive hyperkalemia, hyponatremia, and volume contraction, are other signs that should suggest the diagnosis.

- **Risk factors** – The major risk factors for adrenal hemorrhage or infarction are anticoagulant therapy or an underlying coagulopathy, including the antiphospholipid syndrome, and the postoperative state. In patients treated with an anticoagulant, the results of clotting tests are usually within the therapeutic range, and spontaneous bleeding elsewhere is not evident [96]. (See "[Acquired hemophilia A \(and other acquired coagulation factor inhibitors\)](#)".)

Because adrenal crisis is difficult to recognize clinically, it must be considered whenever these symptoms develop in a patient with one or more risk factors. Without appropriate therapy, shock progresses to coma and death. If the patient survives, adrenal function may rarely return to normal months later [97].

METASTATIC DISEASE

Infiltration of the adrenal glands by metastatic cancer is common, probably because of their rich sinusoidal blood supply. At autopsy, adrenal metastases are found in 40 to 60 percent of patients with disseminated lung or breast cancer, 30 percent of patients with melanoma, and 14 to 20 percent of patients with stomach or colon cancer, but clinically evident adrenal insufficiency is uncommon [110-112]. Similar findings occur with lymphoma [113,114].

The apparent low incidence of clinical adrenal insufficiency in patients with malignant disease is due to the fact that most of the adrenal cortex must be destroyed before hypofunction becomes evident [115,116]. In addition, some of the symptoms of adrenal insufficiency may mistakenly be attributed to cancer. Two small studies have suggested that 20 to 35 percent of patients with bilateral adrenal metastases have at least partial adrenal insufficiency and benefit from glucocorticoid replacement therapy [111,112].

ADVERSE EFFECT OF MEDICATIONS

A number of drugs are associated with adrenal insufficiency by a variety of mechanisms.

- **Checkpoint inhibitor immunotherapy** – Administration of antibodies targeting immune checkpoints (cytotoxic T-lymphocyte-associated antigen 4 [CTLA-4] and programmed death 1 [PD-1]) for cancer therapy have been associated with endocrinopathies. Hypothyroidism is most common, but primary and secondary adrenal insufficiency have also been reported [117]. These are thought to represent autoimmune/autoinflammatory adverse events. (See "[Overview of toxicities associated with immune checkpoint inhibitors](#)", section on '[Endocrinopathies](#)'.)
- **Inhibitors of cortisol biosynthesis** – Several drugs may cause adrenal insufficiency by inhibiting cortisol biosynthesis. They include the anesthetic-sedative drug [etomidate](#) [118], the antimycotic drugs [ketoconazole](#) and [fluconazole](#) [119,120], [metyrapone](#) [121], and the antiparasitic drug [suramin](#), which is also being tested for use in treating prostate cancer [122-124]. These drugs usually do not cause clinically important adrenal insufficiency in patients with normal hypothalamic-pituitary-adrenal function because enzyme inhibition is incomplete and increased corticotropin (ACTH) secretion overrides the pharmacologic blockade. However, patients with limited pituitary or adrenal reserve may develop symptomatic adrenal insufficiency. (See "[Medical therapy of hypercortisolism \(Cushing syndrome\)](#)".)
- **Drugs that accelerate metabolism of cortisol** – There are drugs that accelerate the metabolism of cortisol and most synthetic glucocorticoids by inducing hepatic mixed-function oxygenase enzymes. They include [phenytoin](#) [125,126], barbiturates [125], and [rifampin](#) [125,127,128]. These drugs, therefore, can cause adrenal insufficiency in patients with limited pituitary or adrenal reserve and those with adrenal insufficiency who are receiving glucocorticoid therapy.
- **Mitotane** – The adrenocorticolytic drug [mitotane](#) is used to treat adrenal tumors and to diminish cortisol synthesis in refractory Cushing syndrome. It also accelerates the metabolism of halogenated synthetic steroids such as [dexamethasone](#) and [fludrocortisone](#) [129] and can

provoke adrenal insufficiency in patients with adrenal carcinoma who are receiving steroids of this type. (See "[Treatment of adrenocortical carcinoma](#)", [section on 'Adrenal insufficiency'](#).)

- **Drugs that suppress corticotropin-releasing hormone (CRH) or ACTH production** – This includes glucocorticoids, [megestrol acetate](#) or opioids, or drugs that potentiate glucocorticoid effects by decreasing its metabolism (eg, [ritonavir](#)), are discussed under causes of central (ie, secondary or tertiary) adrenal insufficiency. (See "[Causes of secondary and tertiary adrenal insufficiency in adults](#)".)

SUMMARY

- **Primary adrenal insufficiency** – Adrenal insufficiency is a rare disorder, occurring in up to 144 people per million population. Primary adrenal insufficiency results from disorders of the adrenal cortex and leads to both glucocorticoid and mineralocorticoid deficiency. The most common cause is autoimmune adrenalitis ([table 1](#)). Autoimmune adrenalitis is responsible for 70 to 90 percent of cases, with the remainder caused by other infectious diseases, replacement by metastatic cancer or lymphoma, adrenal hemorrhage or infarction, or drugs. (See '[Introduction](#)' above.)
- **Autoimmune causes** – Autoimmune adrenalitis involves destruction of the cortex by both humoral and cell-mediated immune mechanisms. Antibodies that react with several steroidogenic enzymes (most often 21-hydroxylase) and all three zones of the adrenal cortex are present in the serum in up to 86 percent of patients ([table 2](#)). (See '[Autoimmune adrenalitis](#)' above.)

Up to 50 percent of patients with autoimmune adrenalitis have additional autoimmune disorders; however, adrenal insufficiency is rarely found in patients who present with other autoimmune diseases ([table 5](#)). (See '[Polyglandular autoimmune syndromes](#)' above.)

- **Polyglandular autoimmune syndrome type 1** – Polyglandular autoimmune syndrome type 1 is a rare autosomal recessive disorder caused by mutations in the autoimmune regulator (*AIRE*) gene, which may be important in the selection and generation of regulatory T cells. Hypoparathyroidism or chronic mucocutaneous candidiasis usually appears during childhood or early adolescence, followed by adrenal insufficiency and potentially by other autoimmune diseases. (See '[Type 1 \(monogenic\)](#)' above.)
- **Polyglandular autoimmune syndrome type 2** – Polyglandular autoimmune syndrome type 2 is much more prevalent than type 1 syndrome. Primary adrenal insufficiency is its principal manifestation, but autoimmune thyroid disease and type 1 diabetes mellitus are also common ([table 3](#) and [table 4](#)). Approximately one-half of the cases are familial, with polygenic modes of inheritance. It occurs later than type 1, usually presenting by age 40 years. (See '[Type 2 \(polygenic\)](#)' above.)
- **Infectious causes** – Infectious causes of adrenal insufficiency include tuberculosis, fungal infections, cytomegalovirus and *Mycobacterium avium-intracellulare* (usually in the context of HIV infection), syphilis, and African trypanosomiasis. (See '[Infectious adrenalitis](#)' above.)
- **Bilateral adrenal hemorrhage** – Bilateral adrenal hemorrhage can inhibit adrenal function. This usually occurs in specific settings (see '[Hemorrhagic infarction](#)' above and '[Bilateral adrenal injury](#)' above):
 - Disseminated infection (meningococcus, *Pseudomonas aeruginosa*, *Streptococcus pneumoniae*, *Neisseria gonorrhoeae*, *Escherichia coli*, *Haemophilus influenzae*, and *Staphylococcus aureus*).
 - Clotting abnormalities (anticoagulant drug or heparin therapy or coagulopathy, thromboembolic disease, hypercoagulable states such as antiphospholipid syndrome).
 - Other associations include physical trauma, the postoperative state, sepsis, and severe stress.
- **Metastatic disease** – Metastatic disease with replacement of the cortex of both adrenal glands is a rare cause of adrenal insufficiency. (See '[Metastatic disease](#)' above.)

This is most commonly associated with lung, breast, stomach, or colon cancer, melanoma, and lymphoma.

- **Drugs** – Drugs are an important cause of primary adrenal insufficiency. (See '[Adverse effect of medications](#)' above.)

These include drugs that inhibit cortisol biosynthesis, such as [etomidate](#), [ketoconazole](#), [fluconazole](#), [metyrapone](#), [mitotane](#), and [suramin](#).

Other drugs accelerate the metabolism of cortisol and most synthetic glucocorticoids by inducing hepatic mixed-function oxygenase enzymes (eg, [phenytoin](#), barbiturates, [mitotane](#), and [rifampin](#)). These can cause adrenal insufficiency in patients with limited adrenal reserve who cannot compensate for the accelerated cortisol metabolism.

DISCLOSURE

The views expressed in this topic are those of the author(s) and do not reflect the official views or policy of the United States Government or its components.

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REFERENCES

1. Addison T. On the Constitutional and Local Effects of Disease of the Suprarenal Capsules, Highley, London 1855.
2. Chabre O, Goichot B, Zenaty D, Bertherat J. Group 1. Epidemiology of primary and secondary adrenal insufficiency: Prevalence and incidence, acute adrenal insufficiency, long-term morbidity and mortality. *Ann Endocrinol (Paris)* 2017; 78:490.
3. Betterle C, Presotto F, Furmaniak J. Epidemiology, pathogenesis, and diagnosis of Addison's disease in adults. *J Endocrinol Invest* 2019; 42:1407.
4. Nomura K, Demura H, Saruta T. Addison's disease in Japan: characteristics and changes revealed in a nationwide survey. *Intern Med* 1994; 33:602.
5. Willis AC, Vince FP. The prevalence of Addison's disease in Coventry, UK. *Postgrad Med J* 1997; 73:286.
6. Laureti S, Vecchi L, Santeusano F, Falorni A. Is the prevalence of Addison's disease underestimated? *J Clin Endocrinol Metab* 1999; 84:1762.
7. Erichsen MM, Løvås K, Skinningsrud B, et al. Clinical, immunological, and genetic features of autoimmune primary adrenal insufficiency: observations from a Norwegian registry. *J Clin Endocrinol Metab* 2009; 94:4882.
8. Olafsson AS, Sigurjonsdottir HA. INCREASING PREVALENCE OF ADDISON DISEASE: RESULTS FROM A NATIONWIDE STUDY. *Endocr Pract* 2016; 22:30.
9. Spinner MW, Blizzard RM, Childs B. Clinical and genetic heterogeneity in idiopathic Addison's disease and hypoparathyroidism. *J Clin Endocrinol Metab* 1968; 28:795.
10. Husebye ES, Løvås K. Immunology of Addison's disease and premature ovarian failure. *Endocrinol Metab Clin North Am* 2009; 38:389.
11. Miller WL. Molecular biology of steroid hormone synthesis. *Endocr Rev* 1988; 9:295.
12. Neufeld M, Maclaren NK, Blizzard RM. Two types of autoimmune Addison's disease associated with different polyglandular autoimmune (PGA) syndromes. *Medicine (Baltimore)* 1981; 60:355.
13. Farid NR, Bear JC. The human major histocompatibility complex and endocrine disease. *Endocr Rev* 1981; 2:50.
14. Maclaren NK, Riley WJ. Thyroid, gastric, and adrenal autoimmunities associated with insulin-dependent diabetes mellitus. *Diabetes Care* 1985; 8 Suppl 1:34.
15. Yu L, Brewer KW, Gates S, et al. DRB1*04 and DQ alleles: expression of 21-hydroxylase autoantibodies and risk of progression to Addison's disease. *J Clin Endocrinol Metab* 1999; 84:328.
16. Partanen J, Peterson P, Westman P, et al. Major histocompatibility complex class II and III in Addison's disease. MHC alleles do not predict autoantibody specificity and 21-hydroxylase gene polymorphism has no independent role in disease susceptibility. *Hum Immunol* 1994; 41:135.
17. Badenhop K, Walfish PG, Rau H, et al. Susceptibility and resistance alleles of human leukocyte antigen (HLA) DQA1 and HLA DQB1 are shared in endocrine autoimmune disease. *J Clin Endocrinol Metab* 1995; 80:2112.
18. Ghaderi M, Gambelunghe G, Tortoioli C, et al. MHC2TA single nucleotide polymorphism and genetic risk for autoimmune adrenal insufficiency. *J Clin Endocrinol Metab* 2006; 91:4107.
19. Gambelunghe G, Falorni A, Ghaderi M, et al. Microsatellite polymorphism of the MHC class I chain-related (MIC-A and MIC-B) genes marks the risk for autoimmune Addison's disease. *J Clin Endocrinol Metab* 1999; 84:3701.
20. Baker PR, Baschal EE, Fain PR, et al. Dominant suppression of Addison's disease associated with HLA-B15. *J Clin Endocrinol Metab* 2011; 96:2154.
21. Betterle C, Scalici C, Presotto F, et al. The natural history of adrenal function in autoimmune patients with adrenal autoantibodies. *J Endocrinol* 1988; 117:467.
22. Saenger P, Levine LS, Irvine WJ, et al. Progressive adrenal failure in polyglandular autoimmune disease. *J Clin Endocrinol Metab* 1982; 54:863.
23. Ahonen P, Miettinen A, Perheentupa J. Adrenal and steroidal cell antibodies in patients with autoimmune polyglandular disease type I and risk of adrenocortical and ovarian failure. *J Clin Endocrinol Metab* 1987; 64:494.
24. Baker PR, Nanduri P, Gottlieb PA, et al. Predicting the onset of Addison's disease: ACTH, renin, cortisol and 21-hydroxylase autoantibodies. *Clin Endocrinol (Oxf)* 2012; 76:617.
25. Ahonen P, Myllärniemi S, Sipilä I, Perheentupa J. Clinical variation of autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy (APECED) in a series of 68 patients. *N Engl J Med* 1990; 322:1829.